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Artificial Intelligence (AI) Action Plan

In the interests of sustaining America's global leadership in artificial intelligence, Paragon Health Institute recommends multiple policy actions to be incorporated into the administration's Artificial Intelligence (AI) Action Plan. These recommendations, while ostensibly healthcare-centric, are of strategic value far beyond that sphere. Its cost reduction vision prevents public and private healthcare expenditures from reaching a level where they jeopardize American's AI primacy by consuming financial resources that could otherwise be deployed in the technology's development and implementation. To understand the logic behind this proposal, it is necessary to review the current state of American healthcare expenditures and some of the variables behind it.

As far back as 2009, the Social Security Advisory Board warned the nation's healthcare cost trajectory as "unsustainable" and "perhaps the most significant threat to the long-term economic security of workers and retirees."¹ The Centers for Medicare & Medicaid Services (CMS) reported that U.S. health expenditures grew by 7.5 percent in 2023 (the most recent annual reporting) as compared to 2022.² This increase raised the nation's spending hundreds of billions of dollars from \$4.53 trillion to 4.87 trillion, pushing its share of Gross Domestic Product (GDP) from 17.4 percent to 17.6 percent.³ For comparison, CMS estimated healthcare spending to be 6.9 percent of GDP in 1970⁴ and per capita spending is almost twice that of comparable nations.⁵ A major

¹ Social Security Advisory Board, The Unsustainable Cost of Health Care, September 8, 2009, https://s3-us-gov-west-1.amazonaws.com/cg-778536a2-e58c-44f1-9173-29749804ec54/uploads/2023/12/2009-TheUnsustainableCostofHealthCare_2009.pdf

² See the Centers for Medicare and Medicaid Services' historical national health expenditure data at <https://www.cms.gov/data-research/statistics-trends-and-reports/national-health-expenditure-data/historical>

³ See CMS' "NHE Summary, including share of GDP, CY 1960-2023" files linked from <https://www.cms.gov/data-research/statistics-trends-and-reports/national-health-expenditure-data/historical>

⁴ Ibid.

⁵ Emma Wager et al., "How Does Health Spending in the U.S. Compare to Other Countries?" Peterson-KFF Health System Tracker, (January 23, 2024),

contributing factor are the labor costs in the United States, The average primary care physicians (PCPs) made \$277,000 in 2023.⁶ When specialists are considered along with PCPs, the average American physician earns \$363,000 annually.⁷ When compared to European countries, the average U.S. physician earnings are 120 percent higher than the average in Germany, 189 percent higher than in the United Kingdom, and 278 percent higher than the average for France.⁸ U.S. physician productivity does not appear to compensate for its international salary disparities. In fact, the average American physician sees fewer patients per week than physicians in German, France, and Spain, though they do see significantly more than in the United Kingdom.⁹

This cost burden Americans bear for higher-expense physicians is inflated further by consolidation among health systems across the country. Between 1998 and 2021, 1,887 hospital mergers were announced.¹⁰ A 2023 study from Harvard and the National Bureau of Economic Research found physician services provided by large consolidated health systems priced 12-26 percent more than at independent physician practices, while hospital services averaged 31 percent more in consolidated systems than independent hospitals.¹¹

Collectively these costs become all the more problematic when viewed against the January 2025 report from the U.S. Treasury Department that indicated the nation has an estimated \$36.22 trillion gross national debt,¹² which does not include a far higher

<https://www.healthsystemtracker.org/chart-collection/health-spending-u-s-compare-countries/>

⁶ Jon McKenna, "Medscape Physician Compensation Report 2024: Bigger Checks, Yet Doctors Still See an Underpaid Profession," MedScape, April 12, 2024, <https://www.medscape.com/slideshow/2024-compensation-overview-6017073>

⁷ Ibid.

⁸ Study based on 2022 data and monetary exchange rates. Jon McKenna, "Do US Doctors Have It Better? Medscape International Physician Compensation Report 2023," MedScape, October 11, 2023, <https://www.medscape.com/slideshow/2023-us-vs-global-compensation-report-6016711>

⁹ Ibid.

¹⁰ Hoag Levins, "Hospital Consolidation Continues to Boost Costs, Narrow Access, and Impact Care Quality," Leonard Davis Institute of Health Economics at the University of Pennsylvania, January 19, 2023, <https://ldi.upenn.edu/our-work/research-updates/hospital-consolidation-continues-to-boost-costs-narrow-access-and-impact-care-quality/>

¹¹ Jake Miller, "Care Costs More in Consolidated Health Systems." Harvard Medical School, January 24, 2023, <https://hms.harvard.edu/news/care-costs-more-consolidated-health-systems>

¹² Joint Economic Committee, "Monthly Debt Update," United States Congress, February 7, 2025, <https://www.jec.senate.gov/public/vendor/accounts/JEC-R/debt/mdu/Monthly%20Debt%20Update%20February%202025.pdf>

amount in liabilities and unfunded obligations. America's outsized healthcare spending only exacerbates the nation's deficit spending and debt servicing expenses. Thus, to abide by [Executive Order 13859](#)'s agencies directive to promote sustained investment in AI R&D that contribute to economic and national security, the government needs an initiative that intentionally explores AI for healthcare expenditure reduction lest the technology become one more incremental cost to a prohibitively expensive American health system. AI itself is equipped to remedy industry overspending in ways competing tools of cost control cannot. The range of its capabilities include:

- Reasoning and decision-making that can exceed human performance
- Natural language processing capable of understanding written or verbal information
- Agentic automation capabilities that can initiate tasks and select optimal execution methods based on user-supplied objectives
- Complex pattern detection that can identify the early manifestations of disease within medical images
- Models that can predict future disease risk
- Models whose functionality can improve after deployment based on continuing use and related data analysis
- Faster execution than the same task performed by a clinician
- Task scaling at lower marginal costs than the same tasks performed by a clinician

Unfortunately these AI capabilities, while individually valuable, cannot deliver lower cost healthcare within a piecemeal implementation. Significant obstacles exist between provider-side savings and the transmission of those savings to Americans.¹³ One of the most important of these barriers is the pre-determined payment rates that health insurance plans negotiate with health care providers. This network payment model does not dynamically adjust payments downward based on localized efficiencies. Likewise, providers have no incentive to lobby insurance plans for less compensation in circumstances where AI reduces their labor costs. The same incentive structure prevents these systems from purposely investing in technology that would result in the reduction their payment per service. Negotiating future billing increases to cover the technology costs for AI-assisted services, instead, is the most probable scenario for providers. They will seek compensation for both the costs of AI systems and the overhead involved in its deployment.

In light of these payment challenges, the effort to reduce American healthcare costs through AI needs a policy that explicitly champions the effort as well as selecting an appropriate platform that will serve as a testbed to empirically validate the effort's cost reductions. It is actually the platform, and not the AI scheme, that is critical for this policy

¹³ For a fuller discussion, see Kev Coleman, "Lowering Health Care Costs Through AI: The Possibilities and Barriers," Paragon Health Institute, July 2024, <https://paragoninstitute.org/private-health/lowering-health-care-costs-through-ai-the-possibilities-and-barriers/>

to be a success for reasons that will be explained later in this document. With respect to AI, there is an intuitive methodology for discovering areas where the technology can lower costs albeit one that requires arduous analysis, problem solving, and creativity. The process begins with a decomposition of a statistically relevant claims database (with at least five years of historical data) into healthcare procedures/services sorted from the greatest annual expenditures to the least. For the claims database to be relevant, it must be of a scale where the healthcare utilization trends can be generalized for the nation and the geographic distribution is sufficient to reflect regional variations in population health.

The decomposition can then be reviewed by a joint body of clinicians and AI developers to map the highest expenditure categories to known AI medical devices and healthcare functionality as well as areas where AI can be confidently extrapolated as a solution even if the implementation has not yet taken place. With this mapping completed, the body can then evaluate for each opportunity whether AI can:

- Reduce the existing labor effort (and attendant costs) expended by a primary care physician or specialist to perform the task
- Empower lower cost labor (e.g. nurse practitioners, physician assistants, technicians) to perform the task ordinarily performed by a primary care physician or specialist
- Eliminate the labor ordinarily performed by a healthcare worker through autonomous function inside or outside a clinical setting*
- Improve diagnosis (or patterns suggestive of a diagnosis) and/or treatment determinations to reduce waste due to misdiagnosis or suboptimal therapy and medication choice
- Improve patient-dependent adherence to recommended medication usage, annual exams, vaccine/immunization schedules, etc.

** This determination for outside a clinical setting may involve questions of whether one or more peripheral devices can be economically rented or provided to a patient for use with a computer/mobile device that connects to a remote AI system.*

For a specific health system¹⁴ to serve as an ideal testbed for the spending reduction project that can be extrapolated across the nation, five preconditions that should be satisfied. The health system must:

- Serve at least 3.4 million patients annually
- Serve patients residing across every state in the U.S.
- Own and control its patients' medical claims
- Own and control its patients' EHR data

¹⁴ In this context, a health system is meant to encompass an insurance program as well as its associated healthcare providers.

- Provide access to the protocols detailing patient care directions or how a procedure is performed by a clinician

The 3.4 million patient minimum represents approximately one percent of the nation's population. Given that AI is profoundly dependent on representative datasets to produce results generalizable to the population as a whole, such a large patient base is advisable if not unarguably essential. The advantage of a health system with several million patients is that it is more likely to reflect the complexities of healthcare utilization, patient conditions, and service offerings of the nation. Serving patients in every state is also a major requirement since population health trends vary by region and can influence the outcomes of healthcare, whether AI-enabled or not.

A health system that is simultaneously healthcare provider and insurer provides not only access to medical claims (which can be de-identified for the cost analysis project) but also the Electronic Health Record (EHR) system supporting the patients referenced in the medical claims. The value of the EHR in the cost saving project is twofold. First, its existing data can be scrutinized by AI to identify which therapies and medications are most effective under different patient scenarios. Second, the EHR can be used to track clinical outcomes related to the AI cost-reduction project.

Finally there is the matter of protocols and workflow integration. AI, when applied in the context of a medical service, is governed by a protocol guiding its use. This protocol may extend beyond direct technology interaction for a patient and include training as well as oversight and audits. Likewise, the protocol may operate alongside multiple competing protocols pertaining to the physician and other medical devices and, thus, be integrated within a larger workflow. Having access to, and the prospect to modifying, protocols is a prerequisite for actualizing AI improvements.

The Veterans Health Administration (VHA) health system is the ideal candidate for a healthcare cost reduction project using AI. Not only does the VA have a chief AI director,¹⁵ but it also hosts the National AI Institute which "aims to improve the health and well-being of our Veterans and empower our workforce by harnessing the potential of AI and emerging technologies to deliver safe, effective, and trustworthy solutions."¹⁶ The VHA serves over five million veterans and their families as both insurer and providers, giving it access to EHR data as well as claims data. It operates, as a federal agency, outside state insurance and AI mandates that could impair the experimentation it would need to perform in the context of cost reduction. Another advantage of the VHA

¹⁵ Mitch Mirkin, "VA aims to expand artificial-intelligence research, appoints inaugural AI director," U.S. Department of Veterans Affairs, July 10, 2019,

<https://www.research.va.gov/currents/0719-VA-aims-to-expand-artificial-intelligence-research.cfm>

¹⁶ "VA National Artificial Intelligence Institute," U.S. Department of Veterans Affairs, February 18, 2025, <https://department.va.gov/ai/about/>

is that it does not have a conflict of interest around reducing healthcare expenditures given that it effectively pays for those expenditures. In contrast, for-profit systems must concern themselves with lost revenue and nonprofits will have less excess earnings to redirect into higher salaries or cost-shift accounting practices.¹⁷

The one area where the VHA is a deficient testbed for the expenditure reduction project is fraud, waste, and abuse (FWA) detection within medical claims. One would expect, perhaps naively, that abuses such as upcoding, duplicative and/or unitemized billings would be better investigated in programs with commercial providers such as Medicare or Medicaid.

Regulatory Policy Needs

An important policy that should accompany the AI project to reduce healthcare expenditures concerns the possibility of healthcare that is dispossessed of a clinician through AI functionality. Such autonomous care—the delivery of a medical service via a self-service system a consumer uses without clinician assistance—is not unique to AI. For years, pharmacies have offered simplistic blood pressure machines for unassisted consumer use, with some systems having additional functionality such as body mass index calculation or vision evaluation. AI's differentiation is the sophistication of its software and the additional services it enables. Compared to productivity gains and quality improvements, gains from autonomous AI medical services have among the most dramatic prospects for reducing health expenditures. As such, there should be the regulatory possibility of FDA approval (software as a medical device or SaMD) for AI performing a medical function outside the oversight of a clinician when that AI can empirically satisfy three criteria:

- Accuracy levels (or patient outcomes depending on the nature of the application) equal to or exceeding the average rate for clinicians performing the same function
- No amplification of health risks as compared to when the same function is performed by a clinician
- Output communications that are comprehensible and actionable for the patient without the additional explanation from a physician

¹⁷ Marty Makary, "Hospitals that make profits should pay taxes," STAT, April 14, 2024, <https://www.statnews.com/2024/04/14/nonprofit-hospitals-turn-profit-charity-care-tax-exempt-status/>

A fuller discussion of beneficial AI regulatory principles and warnings against misregulation can be found in “[Healthcare AI Regulation: Guidelines for Maintaining Public Safety and Innovation](#).”¹⁸

Intellectual Property Needs

The treatment of Intellectual property, otherwise known as IP, can promote or constrain the rate of invention within an industry. Positively, IP protections preserve the incentives for AI R&D investments inasmuch as a patent prevents competitors with more financial resources from replicating an invention and then leveraging mature distribution channels to dominate the market and starve early-stage competitors. Negatively, these same protections, if misapplied, can stifle advances as well as encourage the use of AI patents as legal weapons to bleed funds away from AI start-ups rather than protect the rights of new products. During the dot-com period of the late 1990s and early 2000s, start-up technology companies faced a surge of infringement lawsuit threats from so-called patent trolls. Making matters worse is that 60 percent of the companies targeted by trolls were small to mid-sized.¹⁹ For small companies, there was the likelihood of limited resources for legal defense and, therefore, an incentive to settle even if the suit had low merit. While patent litigation was not unique to the dot-com industry, complaints arose regarding the quality of many patents that had been issued for software and the same issues now shadow the AI health care market.

Because of the Patent and Trademark Office’s failures to properly review existing software use and convention, some patents were granted for concepts that were vague or inappropriately broad.²⁰ The U.S. Code stipulates that “Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor.”²¹ The “process” dimension of this statement is what typically applies to software patents (including the AI variety) and may include methods of doing business as facilitated by software. Processes, unfortunately, may not be unique inventions so much as intuitive extensions

¹⁸ Kev Coleman, “Healthcare AI Regulation: Guidelines for Maintaining Public Safety and Innovation,” Paragon Health Institute, December 2024, <https://paragoninstitute.org/private-health/healthcare-ai-regulation/>

¹⁹ Max Baucus, “It’s Time for the U.S. to Tackle Patent Trolls,” Harvard Business Review, September 16, 2022, <https://hbr.org/2022/09/its-time-for-the-u-s-to-tackle-patent-trolls>

²⁰ Daniel Nazer, “Why Is the Patent Office So Bad at Reviewing Software Patents?,” Electronic Frontier Foundation, March 17, 2014, <https://www.eff.org/deeplinks/2014/03/why-patent-office-so-bad-reviewing-software-patents>

²¹ U.S. Patent and Trademark Office, “2104 Requirements of 35 U.S.C. 101 [R-07.2022],” <https://www.uspto.gov/web/offices/pac/mpep/s2104.html>

of a programming language's capability or the mathematical models that inform programming's implementation. Pre-existing algorithms constitute the foundation of AI from linear and Cox regressions to decision trees and Naïve Bayes. When this foundation is not properly taken into account, bad patents are granted, and unproductive infringement lawsuits abound.

To facilitate the needed AI innovation to reduce healthcare costs, the government should:

- Increase internal AI expertise within the Patent and Trademark Office
- Allocate appropriate resources to accommodate extensive prior invention searches needed for AI patent claims
- Promote a mild skepticism around process patents without disallowing them altogether, favoring trade secrets as the preferred method of protection when algorithm claims are heavily indebted to existing theory and practice in the AI field
- Protect AI inventions that are materially novel and not logical extrapolations from foundational math and theory in the public domain
- Incentivize self-service medical AI with high-cost savings potential with a faster patent review and approval process

Conclusion

The present administration's vision for leveraging AI to improve government efficiency and human flourishing provides America an opportunity to address the high healthcare expenditures that have plagued the nation for decades. Reducing these expenditures will free up billions for the public and private sectors and allow them to save more and spend in areas that will produce new social benefits. Additionally, should the cost reduction project be successful, it promises to produce a model that can be shared internationally, particularly in areas of low healthcare access where medical bills can result in debt bondage.